

# 1 July 31, 2025 (Class 1)

Online learning is a sequential learning problem

- No concept of training/test data. Data set is not given at the start at once.
- Decisions on the go. No concept of training loss/accuracy.

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**Algorithm 1** Online Learning Framework

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1: for  $t = 1, 2, \dots$  do  
2:   ALG selects action  $a_t \in \mathcal{A}$   
3:   Environment/adversary selects outcome  $y_t \in \mathcal{Y}$  and/or loss function  $l_t \in \mathcal{L}$   
4:   ALG incurs loss  $l(a_t, y_t)$   
5: end for
```

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## 1.1 Learning with Expert Advice

**Algorithm 2:** Learning with Expert Advice Framework

**input:** N experts

**for**  $t = 1, 2, \dots$  **do**

- Each expert provide advice:  $f_{i,t} \in \mathcal{D}$
- ALG selects action  $a_t \in \mathcal{D}$
- Environment/adversary selects an action  $y_t \in \mathcal{Y}$
- ALG incurs loss  $l(a_t, y_t)$

**end for**

## 1.2 Weighted Majority Algorithm

**Algorithm** Weighted Majority Algorithm

**Input:** N experts, learning rate  $\alpha \in [0, 1)$

**Initialize:**  $w_{i,1} = 1 \forall i \in [N]$

**for**  $t = 1, 2, \dots$  **do**

- Input: Expert recommendations  $f_{i,t} \in \{0, 1\}$
- Predict:  $p_t = \mathbb{I}\{\sum_{i:f_{i,t}=1} w_{i,t} \geq \sum_{i:f_{i,t}=0} w_{i,t}\}$
- Observe:  $y_t \in \{0, 1\}$
- Update weights:  $w_{i,t+1} \leftarrow w_{i,t} \cdot \alpha^{\mathbb{I}\{f_{i,t} \neq y_t\}}$

**end for**

We define the potential function as

$$\Phi(t) = \sum_{i=1}^N w_{i,t-1}$$

### 1.2.1 Mistake bound of Weighted Majority Algorithm

We have,

$$\begin{aligned}
\Phi_{t+1} &= \sum_{i:f_{i,t} \neq y_t} w_{i,t} + \sum_{i:f_{i,t} = y_t} w_{i,t} \\
&= \sum_{i:f_{i,t} \neq y_t} \alpha w_{i,t-1} + \sum_{i:f_{i,t} = y_t} w_{i,t-1} \\
&= \alpha \left( \Phi_t - \sum_{i:f_{i,t} = y_t} w_{i,t-1} \right) + \sum_{i:f_{i,t} = y_t} w_{i,t-1} \\
&= \alpha \Phi_t + (1 - \alpha) \sum_{i:f_{i,t} = y_t} w_{i,t-1} \\
&\leq \alpha \Phi_t + (1 - \alpha) \frac{\Phi_t}{2} \quad (\text{total weight of correct experts is at most half of total weight}) \\
&= \left( \frac{1 + \alpha}{2} \right) \Phi_t
\end{aligned}$$

Now, we have  $\Phi_{t+1} \leq \Phi_t, \forall t$ .

Let  $M_t$  be the number of mistakes made by the algorithm up to time  $t$  and  $T$  be the stopping time. We have,

$$\Phi_{T+1} \leq \left( \frac{1 + \alpha}{2} \right)^{M_T} \Phi_1 = \left( \frac{1 + \alpha}{2} \right)^{M_T} N$$

Let  $M_{i,t}$  TODO: complete this

## 2 August 4, 2025 (Class 2)

## 3 August 7, 2025 (Class 3)

### 3.0.1 Convex function

A function  $f : \mathbb{R}^d \rightarrow \mathbb{R}$  is convex if for all  $x_1, x_2 \in \mathbb{R}^d$  and  $\lambda \in [0, 1]$ ,

$$f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2)$$

### 3.0.2 Strictly convex function

A function  $f : \mathbb{R}^d \rightarrow \mathbb{R}$  is strictly convex if for all  $x_1, x_2 \in \mathbb{R}^d$ ,  $x_1 \neq x_2$  and  $\lambda \in (0, 1)$ ,

$$f(\lambda x_1 + (1 - \lambda)x_2) < \lambda f(x_1) + (1 - \lambda)f(x_2)$$

### 3.0.3 strongly convex function

A function  $f : \mathbb{R}^d \rightarrow \mathbb{R}$  is  $\alpha$ -strongly convex if for all  $x_1, x_2 \in \mathbb{R}^d$  and  $\lambda \in [0, 1]$ ,

$$f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2) - \frac{\alpha}{2} \lambda(1 - \lambda) \|x_1 - x_2\|^2$$

### 3.0.4 Equivalent conditions of strongly convex function

Let  $f$  be a differentiable function on a convex set  $X$ . The following statements are equivalent:

- $f$  is  $\alpha$ -strongly convex on  $X$ .
- For all  $x_1, x_2 \in X$ ,  $\lambda \in [0, 1]$ ,

$$f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2) - \frac{\alpha}{2}\lambda(1 - \lambda)\|x_1 - x_2\|^2$$

- For all  $x_1, x_2 \in X$ ,

$$f(x_1) \geq f(x_2) + \nabla f(x_2)^T(x_1 - x_2) + \frac{\alpha}{2}\|x_1 - x_2\|^2$$

- For all  $x_1, x_2 \in X$ ,

$$(\nabla f(x_1) - \nabla f(x_2))^T(x_1 - x_2) \geq \alpha\|x_1 - x_2\|^2$$

- If  $f$  is twice differentiable, then for all  $x \in X$ ,

$$\nabla^2 f(x) \succeq \alpha I$$

- $f(x) - \frac{\alpha}{2}\|x\|^2$  is convex.

### 3.0.5 smooth function

A differentiable function  $f : \mathbb{R}^d \rightarrow \mathbb{R}$  is  $\beta$ -smooth if for all  $x_1, x_2 \in \mathbb{R}^d$ ,

$$f(x_1) \leq f(x_2) + \nabla f(x_2)^T(x_1 - x_2) + \frac{\beta}{2}\|x_1 - x_2\|^2$$

### 3.0.6 exp concave function

A function  $f : \mathbb{R}^d \rightarrow \mathbb{R}$  is  $\alpha$ -exp concave if for all  $x_1, x_2 \in \mathbb{R}^d$  and  $\lambda \in [0, 1]$ ,

$$h(x) = e^{-\alpha f(x)}$$

is concave.

### 3.0.7 Jensen's inequality

Let  $\phi$  be a convex function and  $X$  be a random variable. Then,

$$\phi(\mathbb{E}[X]) \leq \mathbb{E}[\phi(X)]$$

### 3.0.8 Projection operator

The projection operator  $\Pi_{\mathcal{S}}(y)$  is defined as

$$\Pi_{\mathcal{S}}(y) = \arg \min_{x \in \mathcal{S}} \|x - y\|$$

## 4 August 11, 2025 (Class 4)

## 5 August 14, 2025 (Class 5)

### 5.1 Exponential Weights Algorithm (action setting)

**Algorithm** Exponential Weights Algorithm (action setting)

**Input:**  $N$  experts, learning rate  $\eta > 0$ , a convex loss function  $l$

**Initialize:**  $w_{i,1} = 1 \ \forall i \in [N]$

**for**  $t = 1, 2, \dots$  **do**

– Input:  $f_{i,t} \in \mathcal{D}$  (convex set)

– Algorithm takes action:  $a_t = \frac{\sum_{i=1}^N w_{i,t-1} f_{i,t}}{\sum_{i=1}^N w_{i,t-1}}$

– Adversary selects:  $y_t \in \mathcal{Y}$

– Algorithm incurs loss  $l(a_t, y_t)$

– Update weights:  $w_{i,t} \leftarrow w_{i,t-1} \cdot e^{-\eta l(f_{i,t}, y_t)}$

**end for**

We define the best action in hindsight as

$$a^* = \arg \min_{i \in [N]} \sum_{t=1}^T l(f_{i,t}, y_t)$$

Similarly, we define the overall regret as

$$R_T(\text{exp wts}) = \mathbb{E} \sum_{t=1}^T l(a_t, y_t) - \underbrace{\sum_{t=1}^T l(a^*, y_t)}_{\text{benchmark}}$$

#### 5.1.1 An optimal regret bound for Exponential Weights Algorithm for general convex loss functions

For a general convex loss function  $l$  bounded in  $[a, b]$ , the Exponential Weights Algorithm with learning rate  $\eta > 0$  achieves the following regret bound against any sequence of outcomes  $y_1, \dots, y_T$ . The bound is given by,

$$R_T(\text{exp wts}) \leq \frac{\ln(N)}{\eta} + \frac{\eta(b-a)^2 T}{8}.$$

In particular, choosing  $\eta = \sqrt{\frac{8 \ln(N)}{(b-a)^2 T}}$ , we have,

$$R_T(\text{exp wts}) \leq (b-a) \sqrt{\frac{T}{2} \ln(N)}$$

**Proof:**

**Preliminaries:**

We assume that the loss function  $l$  is convex and bounded in  $[a, b]$ .  
We define,

$$\begin{aligned} L_{i,T} &:= \sum_{t=1}^T l(f_{i,t}, y_t) \\ \hat{L}_T &:= \sum_{t=1}^T l(a_t, y_t) \\ w_{i,t} &:= e^{-\eta L_{i,t}} = e^{-\eta \sum_{s=1}^t l(f_{i,s}, y_s)}, \quad \forall t \geq 1 \\ W_t &:= \sum_{i=1}^N w_{i,t} = \sum_{i=1}^N e^{-\eta L_{i,t}}, \quad \forall t \geq 1 \end{aligned}$$

with  $W_0 = N$ .

We have the following form of the Hoeffding's lemma. Let  $X$  be a random variabe with  $a \leq X \leq b$  almost surely, then for any  $s \in \mathbb{R}$ ,

$$\ln \mathbb{E}[e^{sX}] \leq s\mathbb{E}[X] + \frac{s^2(b-a)^2}{8}$$

**Main proof:**

We have,

$$\begin{aligned} \ln \frac{W_n}{W_0} &= \ln \left( \sum_{i=1}^N \exp(-\eta L_{i,n}) \right) - \ln N \\ &\geq \ln \left( \max_{i \in [N]} e^{-\eta L_{i,n}} \right) - \ln N \\ &= -\eta \min_{i \in [N]} L_{i,n} - \ln N \end{aligned}$$

For  $t = 1, \dots, T$ ,

$$\begin{aligned} \ln \frac{W_t}{W_{t-1}} &= \ln \frac{\sum_{i=1}^N e^{-\eta l(f_{i,t}, y_t)} e^{-\eta L_{i,t-1}}}{\sum_{i=1}^N e^{-\eta L_{i,t-1}}} \\ &= \ln \frac{\sum_{i=1}^N w_{i,t-1} e^{-\eta l(f_{i,t}, y_t)}}{\sum_{i=1}^N w_{i,t-1}} \end{aligned}$$

Using Hoeffding's lemma, we have,

$$\begin{aligned} \ln \frac{\sum_{i=1}^N w_{i,t-1} e^{-\eta l(f_{i,t}, y_t)}}{\sum_{i=1}^N w_{i,t-1}} &\leq -\eta \sum_{i=1}^N \left( \frac{w_{i,t-1}}{\sum_{i=1}^N w_{i,t-1}} \right) l(f_{i,t}, y_t) + \frac{\eta^2(b-a)^2}{8} \\ &= -\eta l(a_t, y_t) + \frac{\eta^2(b-a)^2}{8} \quad (\text{by convexity of } l) \\ \implies \ln \frac{W_t}{W_{t-1}} &\leq -\eta l(a_t, y_t) + \frac{\eta^2(b-a)^2}{8} \end{aligned}$$

Now, summing over  $t = 1, \dots, T$ , we have,

$$\begin{aligned} \ln \frac{W_T}{W_0} &= \sum_{t=1}^T \ln \frac{W_t}{W_{t-1}} \\ &\leq -\eta \sum_{t=1}^T l(a_t, y_t) + \frac{\eta^2(b-a)^2 T}{8} \\ \implies \ln \frac{W_T}{W_0} &\leq -\eta \hat{L}_T + \frac{\eta^2(b-a)^2 T}{8} \end{aligned}$$

Using the lower bound on  $\ln \frac{W_T}{W_0}$ , we derived earlier, we have,

$$\hat{L}_T - \min_{i \in [N]} L_{i,T} \leq \frac{\ln(N)}{\eta} + \frac{\eta(b-a)^2 T}{8}$$

Choosing the optimal  $\eta = \sqrt{\frac{8 \ln(N)}{(b-a)^2 T}}$ , we have,

$$R_T(\text{exp wts}) \leq (b-a) \sqrt{\frac{T}{2} \ln(N)}$$

## 6 August 21, 2025 (Class 6)

Consider  $\mathcal{D} = [0, 1]$ , squared loss  $f^n$  and 3 experts  $f_1, f_2, f_3$  with constant predictions  $f_1(t) = 0$ ,  $f_2(t) = 0.5$ ,  $f_3(t) = 1$ , and  $\mathcal{Y} = \{0, 1\}$ .

### 6.1 An optimal regret bound for Exponential Weights Algorithm for exp concave loss functions

Consider  $\eta$ -exp concave loss function, then the regret is bounded by

$$R_T(\text{exp wts}) \leq \frac{\log(N)}{\eta}$$

**Proof:** We define,

$$\begin{aligned} L_{i,T} &:= \sum_{t=1}^T l(f_{i,t}, y_t) \\ \hat{L}_T &:= \mathbb{E} \sum_{t=1}^T l(a_t, y_t) \end{aligned}$$

We have,

$$\begin{aligned}
R_T(\text{exp wts}) &= \mathbb{E} \sum_{t=1}^T l(a_t, y_t) - \min_{i \in [N]} \sum_{t=1}^T l(f_{i,t}, y_t) \\
&= \max_{i \in [N]} \left[ \mathbb{E} \sum_{t=1}^T l(a_t, y_t) - \sum_{t=1}^T l(f_{i,t}, y_t) \right] \\
&= \max_{i \in [N]} [\hat{L}_T - L_{i,T}] \\
&= \log \left[ \max_{i \in [N]} e^{(\hat{L}_T - L_{i,T})} \right] \\
&= \frac{1}{\eta} \log \left[ \max_{i \in [N]} e^{(\eta(\hat{L}_T - L_{i,T}))} \right] \\
&= \frac{1}{\eta} \log \left[ e^{(\eta(\hat{L}_T - L_{i^*,T}))} \right] \quad \text{where } i^* = \arg \max_{i \in [N]} e^{(\hat{L}_T - L_{i,T})} \\
&\leq \frac{1}{\eta} \log \left[ \sum_{i=1}^N e^{(\eta(\hat{L}_T - L_{i,T}))} \right] \\
\Phi(T) &= \frac{1}{\eta} \log \left[ \sum_{i=1}^N e^{(\eta(\hat{L}_T - L_{i,T}))} \right]
\end{aligned}$$

Now, we have,

$$\begin{aligned}
R_T(\text{exp wts}) &\leq \Phi(T) \\
\text{where } \Phi(T) &= \frac{1}{\eta} \log \left[ \sum_{i=1}^N e^{(\eta(\hat{L}_T - L_{i,T}))} \right] \\
\text{and } \Phi(0) &= \frac{\log(N)}{\eta}
\end{aligned}$$

It is sufficient to show  $\Phi(t) \leq \Phi(t-1) \forall t$ .

$$\begin{aligned}
&\sum_{i=1}^N e^{(\eta(\hat{L}_t - L_{i,t}))} \leq \sum_{i=1}^N e^{(\eta(\hat{L}_{t-1} - L_{i,t-1}))} \\
&\sum_{i=1}^N e^{(\eta\hat{L}_{t-1}) + (\eta\hat{L}_t) - (\eta L_{i,t-1}) - (\eta l_{i,t})} \leq \sum_{i=1}^N e^{(\eta\hat{L}_{t-1}) - (\eta L_{i,t-1})} \\
&\Rightarrow \sum_{i=1}^N \left( \frac{e^{-\eta L_{i,t-1}}}{\sum_{i=1}^N e^{-\eta L_{i,t-1}}} \right) e^{-\eta l_{i,t}} \leq e^{\eta\hat{L}_t}
\end{aligned}$$

TODO: Complete this.

## 7 August 25, 2025 (Class 7)

### 7.1 Stock Portfolio Optimization

We have  $N$  stock options, and  $\delta_N$  is the set of all distribution over  $[N]$ . We take initial wealth as 1, without loss of generality.

We define,

$$r_{i,t} = \frac{\text{closing value of stock } i \text{ on day } t}{\text{opening value of stock } i \text{ on day } t}$$

$(P_t)_{t \leq T}$  : Action sequence, the distribution of wealth on day  $t$

$(r_t)_{t \leq T}$  : Market (reward) sequence, the return of each stock on day  $t$

We have the wealth at stopping day  $T$ ,

$$W_T = \prod_{t=1}^T \langle P_t, r_t \rangle$$

We have the max possible wealth as,

$$W_T^* = \max_{P \in \delta_N} \prod_{t=1}^T \langle P_t, r_t \rangle$$

and the best constant distribution as,

$$P^* = \arg \max_{P \in \delta_N} \prod_{t=1}^T \langle P_t, r_t \rangle$$

$$\log \left( \frac{W_T}{W_T^*} \right) = \sum_{i=1}^T \log(\langle P_t, r_t \rangle) - \sum_{i=1}^T \log(\langle P^*, r_t \rangle)$$

$$l(P_t, r_t) = -\log(\langle P_t, r_t \rangle^{1'})$$

## 8 September 1, 2025 (Class 8)

## 9 September 4, 2025 (Class 9)

We have an online convex optimization setting

Let  $\mathcal{K}$  be the set of values  $x_t$  which the algorithm can choose, and  $\mathcal{F}$  be the set of all convex loss functions.

**Algorithm:** Online Convex Optimization

**input:**  $\mathcal{K} :=$  set of values,  $\mathcal{F} :=$  set of convex loss functions

**for**  $t = 1, 2, \dots$  **do**

– Algorithm selects action  $x_t \in \mathcal{K}$

– Environment selects a loss function  $f_t \in \mathcal{F}$

– ALG incurs loss  $f_t(x_t)$

**end for**

Do note, the env only gives  $f_t(x_t)$  and  $\nabla f_t(x_t)$ , not  $f_t$  itself.



## 9.1 Online gradient descent

**Algorithm:** Online Gradient Descent

**input:**  $\mathcal{K} :=$  set of values,  $\mathcal{F} :=$  set of convex loss functions

**for**  $t = 1, 2, \dots$  **do**

– Algorithm selects action  $x_t \in \mathcal{K}$

– Environment selects a loss function  $f_t \in \mathcal{F}$

– ALG incurs loss  $f_t(x_t)$

– Update  $x_{t+1} \leftarrow \Pi_{\mathcal{K}}(x_t - \eta_t \nabla f_t(x_t))$

**end for**

### 9.1.1 Regret guarantee on online gradient descent

We have,

$$R_T(OGD) \leq \frac{3}{2} DG\sqrt{T},$$

where

$$G := \sup_{f \in \mathcal{F}} \sup_{x \in \mathcal{K}} \|\nabla f(x)\|$$

$$D := \sup_{x, x' \in \mathcal{K}} \|x - x'\|_2$$

**Proof:** We define

$$\nabla_t := \nabla f_t(x)$$

Also,  $x^* \in \mathcal{K}$  as

$$x^* = \arg \min_{x \in \mathcal{K}} \sum_{i=1}^T f_i(x).$$

From the first order condition of convexity,

$$\begin{aligned} f_t(x_t) - f_t(x^*) &\leq \nabla_t^T (x_t - x^*) \\ \sum_{t=1}^T f_t(x_t) - \sum_{t=1}^T f_t(x^*) &\leq \sum_{t=1}^T \nabla_t^T (x_t - x^*) \\ \implies R_T(OGD) &\leq \sum_{t=1}^T \nabla_t^T (x_t - x^*). \end{aligned}$$

Now,

$$\begin{aligned} \|x^* - x_{t+1}\| &= \|x^* - \Pi_{\mathcal{K}}(x_t - \eta_t \nabla_t)\| \\ &\leq \|x^* - x_t + \eta_t \nabla_t\|, \quad \text{by Pythagorean theorem} \\ \|x^* - x_{t+1}\|^2 &\leq \|x^* - x_t\|^2 + \eta_t^2 \|\nabla_t\|^2 + 2\eta_t \langle x^* - x_t, \nabla_t \rangle \\ -2\eta_t \langle x^* - x_t, \nabla_t \rangle &\leq \|x^* - x_t\|^2 - \|x^* - x_{t+1}\|^2 + \eta_t^2 \|\nabla_t\|^2 \\ 2 \langle x_t - x^*, \nabla_t \rangle &\leq \frac{1}{\eta_t} (\|x^* - x_t\|^2 - \|x^* - x_{t+1}\|^2) + \eta_t \|\nabla_t\|^2 \end{aligned}$$

From the above two results, we have

$$\begin{aligned}
2R_T(OGD) &\leq \sum_{t=1}^T \left( \frac{\|x^* - x_t\|^2 - \|x^* - x_{t+1}\|^2}{\eta_t} \right) + \sum_{t=1}^T \eta_t \|\nabla_t\|^2 \\
&= \frac{\|x^* - x_1\|^2}{\eta_1} + \sum_{t=2}^T \|x^* - x_t\|^2 \left( \frac{1}{\eta_t} - \frac{1}{\eta_{t-1}} \right) - \frac{\|x^* - x_{T+1}\|^2}{\eta_T} + \sum_{t=1}^T \eta_t \|\nabla_t\|^2 \\
&\leq \frac{\|x^* - x_1\|^2}{\eta_1} - \frac{D^2}{\eta_1} + \frac{D^2}{\eta_T} - \frac{\|x^* - x_{T+1}\|^2}{\eta_T} + G^2 \sum_{t=1}^T \eta_t \\
&\leq \frac{D^2}{\eta_T} + G^2 \sum_{t=1}^T \eta_t
\end{aligned}$$

Putting  $\eta_t = \frac{D}{G\sqrt{t}}$ , we have

$$\begin{aligned}
2R_T(OGD) &\leq D^2 \frac{G\sqrt{T}}{D} + G^2 \frac{D}{G} \sum_{t=1}^T \frac{1}{\sqrt{t}} \\
&\leq DG\sqrt{T} + 2DG\sqrt{T} \\
\implies R_T(OGD) &\leq \frac{3}{2}DG\sqrt{T}
\end{aligned}$$

### 9.1.2 Regret guarantee on online gradient descent for strongly convex loss function

With  $\eta = 1/(\alpha t)$ , we have

$$R_T(OGD) \leq \frac{G^2}{2\alpha} (1 + \log(T))$$

**Proof:** From  $\alpha$ -strong convexity property we have

$$2(f_t(x_t) - f_t(x^*)) \leq 2\nabla_t^T(x_t - x^*) - \alpha\|x_t - x^*\|^2$$

We will use the upper bound on the first term from [previous](#) proof. We have TODO: check and correct this.

$$\begin{aligned}
2R_T(OGD) &= \sum_{t=1}^T 2(f_t(x_t) - f_t(x^*)) \\
&\leq \sum_{t=1}^T \left( \frac{\|x_t - x^*\|^2 - \|x_{t+1} - x^*\|^2}{\eta_t} - \alpha\|x_t - x^*\|^2 \right) + G^2 \sum_{t=1}^T \eta_t - \frac{\|x_{T+1} - x^*\|^2}{\eta_T} \\
&\leq D^2 \underbrace{\sum_{t=1}^T \left( \frac{1}{\eta_t} - \frac{1}{\eta_{t-1}} - \alpha \right)}_{=0} + G^2 \sum_{t=1}^T \eta_t \\
&= \frac{G^2}{\alpha} \sum_{t=1}^T \frac{1}{t} \\
&\leq \frac{G^2}{\alpha} (1 + \log(T))
\end{aligned}$$

## 9.2 Follow the leader (FTL)

**Algorithm:** Follow The Leader (FTL)

**Input:** A convex set of possible actions (decision set)  $\mathcal{K}$ .

**for**  $t = 1, 2, 3, \dots, T$  **do**

1. Play the action  $x_t$  that minimizes the cumulative loss observed so far:

$$x_t := \arg \min_{x \in \mathcal{K}} \sum_{s=1}^{t-1} f_s(x)$$

(Note: For the first round,  $t = 1$ , the sum is empty, so any action  $x_1 \in \mathcal{K}$  can be chosen.)

2. The environment reveals the loss function for the current round,  $f_t$ .

3. The algorithm incurs and observes the loss  $f_t(x_t)$ .

**end for**

### 9.2.1 FTL-BTL lemma

**Lemma 9.1 (FTL-BTL lemma<sup>1</sup>):** *Let  $x_1, x_2, \dots$  be the sequence of points chosen by FTL. Then, for any  $u \in \mathcal{K}$  and for any stopping time  $T$ , we have*

$$\sum_{t=1}^T f_t(x_{t+1}) \leq \sum_{t=1}^T f_t(u)$$

**Proof:** TODO: using induction

TODO: Follow the Regularized Leader For linear loss

## 9.3 Follow the regularized leader (FTRL)

**Remark 9.2:** *The regularizers considered are bounded,  $\alpha$ -strongly convex functions.*

**Algorithm 11:** FTRL Algorithm for general loss function

**Input:** convex set  $\mathcal{K}$ , function class  $\mathcal{F}$ , regularization function  $R$

**Initialize:**  $x_1 = \arg \min_{x \in \mathcal{K}} R(x)$

**for**  $t = 1, 2, \dots$  **do**

- Algorithm plays  $x_t \in \mathcal{K}$
- Environment reveals  $f_t \in \mathcal{F}$
- Algorithm incurs a loss  $f_t(x_t) \in \mathbb{R}$
- Update

$$x_{t+1} := \arg \min_{x \in \mathcal{K}} \left[ \sum_{s=1}^t \langle \nabla_s, x \rangle + \frac{1}{\eta} R(x) \right]$$

**end for**

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<sup>1</sup>Follow the leader, be the leader lemma

### 9.3.1 FTRL with linear loss function and negative entropic regularizer

## 10 September 8, 2025 (Class 10)

### 10.1 Some mathematical preliminaries

**Definition 10.1 (Bregman divergence):** We define the Bregman divergence, with respect to function  $R$ , as

$$B_R(x\|y) = R(x) - R(y) - \langle \nabla R(y), x - y \rangle$$

**Remark 10.2:** For twice differentiable functions, the Bregman divergence is equal to the second derivative at an intermediate point.

**note:** For  $R(x) = \frac{1}{2}\|x\|_{L_2}^2$ , we have

$$B_R(x\|y) = \frac{1}{2}\|x - y\|^2.$$

For  $R(x) = \sum_{i=1}^d x_i \log x_i$ ,  $\sum x_i = 1$  and  $d = 2$ , we have

$$B_R(x\|y) = D_{KL}(x\|y)$$

**Properties of Bregman divergence:**

- It is strictly convex in the first argument
- Non-negative  $B_R(x\|y) \geq 0$ . The proof follows from Taylor's theorem.
- Asymmetric i.e.  $B_R(x\|y) \neq B_R(y\|x)$ .
- Non-convex in the second argument. Example  $R(x) = -\log(x)$  and  $B_R(x\|y) = \log(y) - \log(x) - \frac{x-y}{y}$
- $\frac{\partial B_R(x\|y)}{\partial x} = \nabla R(x) - \nabla R(y)$ .
- Cosine inequality

$$B_R(x\|y) + B_R(y\|z) = B_R(x\|z) + \langle x - y, \nabla R(z) - \nabla R(y) \rangle$$

**Definition 10.3 (Dual norm):** Let  $\|\cdot\|_*$  be a norm on the vector space  $\mathcal{K} \subseteq \mathbb{R}^n$ , then the function  $\|\cdot\|^*$  defined as

$$\|x\|^* = \sup_{y \in \mathcal{K}, \|y\|_* \leq 1} \langle x, y \rangle = \sup_{y \in \mathcal{K}} \left\langle x, \frac{y}{\|y\|_*} \right\rangle,$$

is called a dual norm.

**Definition 10.4 (Norm induced by a symmetric positive definite matrix):** Let  $A \in S_{++}^n$ <sup>2</sup> be a matrix. We define the norm induced by  $A$  as

$$\|x\|_A = \sqrt{x^T A x}$$

---

<sup>2</sup>symmetric positive definite matrix

**Lemma 10.5:** Let  $\|x\|_A$  be a norm where  $A \in S_{++}^n$ . We have the dual norm

$$\|x\|_A^* = \sqrt{x^T A^{-1} x}$$

**Proof:**

## 11 September 11, 2025 (Class 11)

### 11.1 FTRL for general loss functions

**Algorithm:** FTRL Algorithm for general loss function

**Input:** convex set  $\mathcal{K}$ , function class  $\mathcal{F}$ , regularization function  $R$

**Initialize:**  $x_1 = \arg \min_{x \in \mathcal{K}} R(x)$

**for**  $t = 1, 2, \dots$  **do**

- Algorithm plays  $x_t \in \mathcal{K}$
- Environment reveals  $f_t \in \mathcal{F}$
- Algorithm incurs a loss  $f_t(x_t) \in \mathbb{R}$
- Update

$$x_{t+1} := \arg \min_{x \in \mathcal{K}} \left[ \sum_{s=1}^t \langle \nabla_s, x \rangle + \frac{1}{\eta} R(x) \right]$$

**end for**

**Lemma 11.1:** Let  $x_1, x_2, \dots$  be a sequence of points chosen by FTRL, then for any  $u \in \mathcal{K}$  and  $t \geq 1$ , we have

$$\sum_{s=1}^t f_s(x_{s+1}) - \sum_{s=1}^t f_s(u) \leq \frac{R(u) - R(x_1)}{\eta}$$

:

#### 11.1.1 Regret upper bound of FTRL

We have the regret upper bound as

$$R_T(\text{FTRL}) \leq \eta \sum_{t=1}^T (\|\nabla_t\|_t^*)^2 + \frac{R(u) - R(x_1)}{\eta}$$

**Proof:** TODO

## 12 September 15, 2025 (Class 12)

### 12.1 Online Mirror Descent (OMD)

**Algorithm: Online Mirror Descent (OMD)**

**Input:** convex set  $\mathcal{K}$ , function class  $\mathcal{F}$ , regularization function  $R$

**Initialize:**  $x_1 = \arg \min_{x \in \mathcal{K}} R(x)$  and  $y_1$  such that  $\nabla R(y_1) = 0$  ;

**for**  $t = 1, 2, \dots$  **do**

- Algorithm plays  $x_t \in \mathcal{K}$  ;
- Environment reveals  $f_t \in \mathcal{F}$  ;
- Algorithm incurs a loss  $f_t(x_t) \in \mathbb{R}$  ;
- Update

$y_{t+1}$  such that  $\nabla R(y_{t+1}) = \nabla R(y_t) - \eta \nabla_t$  [Lazy version]

$y_{t+1}$  such that  $\nabla R(y_{t+1}) = \nabla R(x_t) - \eta \nabla_t$  [Agile version]

- $x_{t+1} = \arg \min_{x \in \mathcal{K}} B_R(x \| y_{t+1})$

**end**

#### 12.1.1 Equivalence of lazy OMD and FTRL

**Theorem 12.1 (Equivalence of lazy OMD and FTRL):** Let  $\mathcal{F}$  be the class of loss functions. Then the lazy OMD and FTRL algorithms play the same points,

$$\arg \min_{x \in \mathcal{K}} (B_R(x \| y)) = \arg \min_{x \in \mathcal{K}} \left( \eta \sum_{s=1}^{t-1} \nabla_s^T x + R(x) \right)$$

**Proof:** TODO

#### 12.1.2 Agile OMD

**Theorem 12.2:** Let  $\mathcal{K}$  be a convex set and  $x' = \Pi_{\mathcal{K}} B_R(x \| y)$  be a Bregman projection of some  $y \in \mathbb{R}^n$  on  $\mathcal{K}$  and  $u \in \mathcal{K}$ . Then

$$B_R(y \| u) \geq B_R(y \| x) + B_R(x \| u)$$

**Theorem 12.3 (Regret bound of Agile OMD):** For  $\forall u \in \mathcal{K}$ , we have

$$R_T(OMD) \leq \frac{\eta}{2} \sum_{t=1}^T (\|\nabla\|_t^*)^2 + \frac{D_R^2}{\eta}$$

## 13 September 22, 2025 (Class 13)

### 13.1 Bandits setting

**Algorithm:** Multi-Armed Bandits setting  
**input:**  $n$  := number of arms  
**for**  $t = 1, 2, \dots$  **do**  
    – Algorithm picks an arm  $i_t \in [n]$   
    – Environment reveals loss  $l_{i_t, t} \in [0, 1]$   
**end for**

### 13.2 Naive application of online gradient descent for bandit setting

**Algorithm:** Online gradient descent for bandit setting  
**input:**  $n$  := number of arms,  $\delta \in (0, 1)$   
**for**  $t = 1, 2, \dots$  **do**  
     $b_t \sim \text{Bern}(\delta)$   
    **if**  $b_t = 1$  **then**  
        –  $i_t \sim \text{Unif}([n])$  Exploration  
        –  $\hat{\ell}_{i_t, t} = \begin{cases} \frac{n}{\delta} \ell_{i_t, t} & \text{if } i = i_t \\ 0 & \text{otherwise} \end{cases}$   
        – Set  $\hat{f}_t = \hat{\ell}_{i_t, t} \cdot x_{i_t}$  for all  $i \in [n]$ , i.e.  $\hat{f}_t = \langle \hat{\ell}_t, x \rangle$   
        –  $x_{t+1} = \text{OGD}(\hat{f}_1, \hat{f}_2, \dots, \hat{f}_t)$   
    **end if**  
    **else if**  $b_t = 0$  **then**  
        –  $i_t \sim x_t$  Exploitation  
        –  $x_{t+1} = x_t, \hat{\ell}_t = 0$  and  $\hat{f}_t = 0$   
    **end if**  
**end for**

**Lemma 13.1:**  $\mathbb{E}[l_{i_t, t}] \leq \mathbb{E}[\langle \hat{l}_t, x_t \rangle] + \delta$

**Proof:**

#### 13.2.1 Regret upper bound of OGD-MAB

We have,

$$\mathbb{E}[R_T(\text{OGD} - \text{MAB})] \leq 4(nT)^{2/3},$$

where we chose  $\delta = n^{2/3}T^{-1/3}$

**Proof:**

$$\begin{aligned}
\mathbb{E}[R_T(OGD - MAB)] &= \mathbb{E} \left[ \sum_{t=1}^T l_{i_t, t} - \sum_{t=1}^T l_{i^*, t} \right] \\
&= \mathbb{E} \left[ \sum_{t=1}^T l_{i_t, t} - \sum_{t=1}^T \hat{l}_{i^*, t} \right] \\
&\leq \mathbb{E} \left[ \sum_{t=1}^T \langle l_t, x_t \rangle - \sum_{t=1}^T \hat{l}_{i^*, t} \right] + \delta T \\
&= \mathbb{E}[R_T(OGD)] + \delta T \\
&\leq \frac{3}{2} GD \sqrt{\delta T} + \delta T, \quad G = \sup \|\hat{l}\| \leq n/\delta, D \leq 2 \\
&\leq \frac{3n}{\delta} \sqrt{\delta T} + \delta T \\
&\leq \frac{3n}{\delta} \sqrt{n}
\end{aligned}$$

## 14 September 25, 2025 (Class 14)

### 14.1 EXP3- $\gamma$

**Algorithm : EXP3- $\gamma$**

**Input:** Number of arms  $n, \gamma \in (0, 1)$

**Initialize:**  $w_{i,1} = 1$  for all  $i \in [n]$

**for**  $t = 1, 2, \dots$  **do**

- $p_{i,t} = (1 - \gamma) \frac{w_{i,t}}{\sum_{j=1}^n w_{j,t}} + \frac{\gamma}{n}$
- Draw  $i_t \sim p_t$  where  $p_t = (p_{1,t} \ p_{2,t} \ \dots \ p_{n,t})^T$
- Observe  $r_{i_t, t}$ , the reward at time  $t$  (from arm  $i_t$ )
- $\hat{r}_{i,t} = \mathbb{I}\{i_t = i\} \frac{r_{i_t, t}}{p_{i_t, t}}$
- $w_{i,t+1} = w_{i,t} \cdot e^{\eta \hat{r}_{i,t}}$

**end**

**Theorem 14.1:** For any  $\gamma \in (0, 1)$ , any reward sequence  $(r_t)_{t \geq 1}$  with  $r_t \in [0, 1]^n$ , we have

$$\mathbb{E}[R_T(EXP3 - \gamma)] = \max_i \sum_{t=1}^T r_{i,t} - \mathbb{E} \left[ \sum_{t=1}^T r_{i_t, t} \right] \leq 2\gamma T + \frac{n \log(n)}{\gamma}.$$

Furthermore, taking  $\gamma = \sqrt{\frac{n \log(n)}{2T}}$ ,

$$\mathbb{E}[R_T(EXP3 - \gamma)] \leq 2\sqrt{2Tn \log(n)}$$

**Proof:** Let  $W_t = \sum_{i=1}^n w_{i,t}$ .



Upper bounding  $\log(W_{T+1}/W_1)$ ,

$$\begin{aligned}
\frac{W_{t+1}}{W_t} &= \sum_{i=1}^n \frac{w_{i,t+1}}{W_t} = \sum_{i=1}^n \left( \frac{w_{i,t}}{W_t} \right) e^{\gamma \hat{r}_{i,t}/n} \\
&= \sum_{i=1}^n \frac{p_{i,t} - \gamma/n}{1 - \gamma} \cdot \exp\left(\frac{\gamma \hat{r}_{i,t}}{n}\right) \\
&\leq \sum_{i=1}^n \frac{p_{i,t} - \gamma/n}{1 - \gamma} \cdot \exp\left(\frac{\gamma \hat{r}_{i,t}}{n}\right) \\
&\leq \sum_{i=1}^n \frac{p_{i,t} - \gamma/n}{1 - \gamma} \left( 1 + \frac{\gamma}{n} \hat{r}_{i,t} + \left(\frac{\gamma}{n}\right)^2 \hat{r}_{i,t}^2 \right) \\
&\quad \text{(Using } e^x \leq 1 + x + x^2 \text{ for all } x \in [0, 1]) \\
&\leq \frac{1}{1 - \gamma} \sum_{i=1}^n p_{i,t} \left( 1 + \frac{\gamma}{n} \hat{r}_{i,t} + \left(\frac{\gamma}{n}\right)^2 \hat{r}_{i,t}^2 \right) \\
&\leq \frac{1}{1 - \gamma} \left( \underbrace{\sum_{i=1}^n p_{i,t}}_{=1} + \frac{\gamma}{n} \underbrace{\sum_{i=1}^n p_{i,t} \hat{r}_{i,t}}_{=r_{i_t,t}} + \left(\frac{\gamma}{n}\right)^2 \sum_{i=1}^n \underbrace{p_{i,t} \hat{r}_{i,t}}_{\leq 1} \hat{r}_{i,t} \right) \\
\log\left(\frac{W_{t+1}}{W_t}\right) &\leq \frac{1}{1 - \gamma} \left( \frac{\gamma}{n} r_{i_t,t} + \left(\frac{\gamma}{n}\right)^2 \sum_{i=1}^n \hat{r}_{i,t} \right), \quad \text{using } \log(1 + x) \leq x \text{ and } \frac{1}{1 - \gamma} \geq 1 \\
\Rightarrow \log\left(\frac{W_{T+1}}{W_1}\right) &\leq \frac{1}{1 - \gamma} \left( \frac{\gamma}{n} \sum_{t=1}^T r_{i_t,t} + \left(\frac{\gamma}{n}\right)^2 \sum_{t=1}^T \sum_{i=1}^n \hat{r}_{i,t} \right), \quad \text{summing from } t = 1 \text{ to } T
\end{aligned}$$

Lower bounding  $\log(W_{T+1}/W_1)$ ,

$$\begin{aligned}
\log\left(\frac{W_{T+1}}{W_1}\right) &\geq \log\left(\frac{w_{i^*,T+1}}{W_1}\right) \\
&= \log(e^{\gamma/n \sum_{t=1}^T \hat{r}_{i^*,t}}) - \log(n), \quad \because W_1 = n \\
&= \frac{\gamma}{n} \sum_{t=1}^T \hat{r}_{i^*,t} - \log(n)
\end{aligned}$$

Let  $i^*$  be the best arm in hindsight. Combining the above upper and lower bounds for  $i^*$ ,

$$\begin{aligned}
\frac{\gamma}{n} \sum_{t=1}^T \hat{r}_{i^*,t} - \log(n) &\leq \frac{1}{1-\gamma} \left( \frac{\gamma}{n} \sum_{t=1}^T r_{i_t,t} + \left(\frac{\gamma}{n}\right)^2 \sum_{t=1}^T \sum_{i=1}^n \hat{r}_{i,t} \right) \\
\frac{\gamma}{n} \sum_{t=1}^T r_{i^*,t} - \log(n) &\leq \frac{1}{1-\gamma} \left( \frac{\gamma}{n} \mathbb{E} \left[ \sum_{t=1}^T r_{i_t,t} \right] + \left(\frac{\gamma}{n}\right)^2 \sum_{t=1}^T \sum_{i=1}^n r_{i,t} \right), \quad \because \mathbb{E}[\hat{r}_{i,t}] = r_{i,t} \\
(1-\gamma) \sum_{t=1}^T r_{i^*,t} - \mathbb{E} \left[ \sum_{t=1}^T r_{i_t,t} \right] &\leq \frac{(1-\gamma)n \log(n)}{\gamma} + \underbrace{\left(\frac{\gamma}{n}\right) \sum_{t=1}^T \sum_{i=1}^n r_{i,t}}_{\leq n} \\
\sum_{t=1}^T r_{i^*,t} - \mathbb{E} \left[ \sum_{t=1}^T r_{i_t,t} \right] &\leq \frac{(1-\gamma)n \log(n)}{\gamma} + \gamma T + \gamma \sum_{t=1}^T r_{i^*,t} \\
\mathbb{E}[R_T(EXP3 - \gamma)] &\leq \frac{n \log(n)}{\gamma} + 2\gamma T, \quad \because r_{i^*,t} \leq 1
\end{aligned}$$

Taking  $\gamma = \sqrt{\frac{n \log(n)}{2T}}$ ,

$$\mathbb{E}[R_T(EXP3 - \gamma)] \leq 2\sqrt{2Tn \log(n)}$$

**Remark 14.2:** Using a tighter bound of  $e^x$  in the upper bounding gives a better constant.

## 15 October 6, 2025 (Class 15)

### 15.1 Stochastic Multi Armed Bandits

**Lemma 15.1:** Let  $N_{i,T}$  be the number of times the arm  $i$  has been pulled till time  $T$  by an algorithm and let  $\Delta_i = \mu_{i^*} - \mu_i$  be the suboptimality gap. Then,

$$R_T(Alg) = \sum_{i=1}^N \Delta_i \mathbb{E}[N_{i,T}]$$

**Proof:**

$$N_{i,T} = \sum_{t=1}^T \mathbb{I}(i_t = i)$$

We have,

$$\begin{aligned}
\sum_{i=1}^N \Delta_i \mathbb{E}[N_{i:T}] &= \mathbb{E} \left[ \sum_{i=1}^N (\mu_{i^*} - \mu_i) \left( \sum_{t=1}^T \mathbb{I}(i_t = i) \right) \right] \\
&= \mathbb{E} \left[ \sum_{i=1}^N \mu_{i^*} \sum_{t=1}^T \mathbb{I}(i_t = i) - \sum_{i=1}^N \mu_i \sum_{t=1}^T \mathbb{I}(i_t = i) \right] \\
&= \mu_{i^*} \left( \sum_{t=1}^T \sum_{i=1}^N \mathbb{E}[\mathbb{I}(i_t = i)] \right) - \mathbb{E} \left[ \sum_{t=1}^T \sum_{i=1}^N \mu_i \mathbb{I}(i_t = i) \right] \\
&= \mu_{i^*} T - \sum_{t=1}^T \mu_{i_t}, \quad \because \sum_{i=1}^N \mathbb{E}[\mathbb{I}(i_t = i)] = 1 \\
\sum_{i=1}^N \Delta_i \mathbb{E}[N_{i:T}] &= R_T(ALG)
\end{aligned}$$

## 16 October 9, 2025 (Class 16)

### 16.1 Exploration separate algorithm

We have, from the [Hoeffding's Inequality],

$$\begin{aligned}
Pr\{|\hat{\mu}_{i,t} - \mu_i| \geq \varepsilon_{i,t}\} &\leq 2 \exp(-2N_{i,t}\varepsilon_{i,t}^2) \\
Pr\{\hat{\mu}_{i,t} - \mu_i \geq \varepsilon_{i,t}\} &\leq \exp(-2N_{i,t}\varepsilon_{i,t}^2)
\end{aligned}$$

where

$$\begin{aligned}
N_{i,t} &:= \text{times arm}(i) \text{ is pulled till time}(t) \\
\hat{\mu}_{i,t} &:= \text{emperical average till time } (t).
\end{aligned}$$

We define

$$\begin{aligned}
\delta &:= \exp(-2N_{i,t}\varepsilon_{i,t}^2) \\
\Rightarrow \varepsilon_{i,t} &= \sqrt{\frac{\log(1/\delta)}{2N_{i,t}}}
\end{aligned}$$

#### 16.1.1 Regret analysis of exploration separate algorithm

$$\begin{aligned}
R_T(ALG) &= \sum_{i=1}^N \Delta_i \mathbb{E}[N_{i,t}] \\
&= \sum_{i=1}^N \Delta_i \mathbb{E}[N_{i,t'}] + \sum_{i=1}^N \Delta_i \mathbb{E}[N_{i,t'+1:T}]
\end{aligned}$$

We have,

$$\underbrace{j}_{r.v.} = \arg \max_{i \in [N]} \hat{\mu}_{i,t'}$$

$$i^* = \arg \max_{i \in [N]} \mu_i$$

$$\delta = 1/T^4 \implies \varepsilon_{i,t'} = \sqrt{\frac{2 \log T}{N_{i,t'}}}$$

Using the above,

$$\begin{aligned} R_T(\text{ALG}) &= \sum_{i=1}^N \Delta_i \frac{\alpha T}{N} + (T - t') \sum_{i=1}^N \mathbb{I}(i = j) \nabla_j \\ &\leq \alpha T + T \sum_{i=1}^N (\mu_{i^*} - \mu_i) \mathbb{I}(i = j), \quad (\Delta_i < 1; t' > 0) \end{aligned}$$

Now, we have

$$\begin{aligned} \mu_j &\geq \hat{\mu}_j - \varepsilon_j \quad \text{w.p. at least } (1 - 1/T^4) \\ \mu_{i^*} - \mu_j &\leq \mu_{i^*} - \hat{\mu}_j + \varepsilon_j \end{aligned}$$

TODO: complete this.

### 16.1.2 Instance independent (distribution-free) regret

The instance independent (distribution-free) regret bound for the exploration-separate algorithm is

$$R_T(\text{ALG}) \leq \alpha T + T \cdot \mathbb{P}(j \neq i^*)$$

where  $\alpha T$  is the regret incurred during the exploration phase, and the second term is the regret incurred during the exploitation phase if the empirically best arm  $j$  is not the true best arm  $i^*$ .

Using Hoeffding's inequality, with probability at least  $1 - 1/T^4$ , the empirical mean of each arm is within  $\varepsilon_{i,t'}$  of its true mean. Thus, the probability of choosing a suboptimal arm in the exploitation phase is at most  $N/T^4$ , so the regret in the exploitation phase is at most  $T \cdot N/T^4 = N/T^3$ , which is negligible for large  $T$ .

Therefore, the total regret is

$$R_T(\text{ALG}) \leq \alpha T + o(1)$$

### 16.1.3 Instance dependent (distribution-dependent) regret

The instance dependent regret bound depends on the suboptimality gaps  $\Delta_i = \mu_{i^*} - \mu_i$ . After the exploration phase, with high probability, the empirically best arm is the true best arm. Thus, the regret is dominated by the exploration phase:

$$R_T(\text{ALG}) \leq \sum_{i: \Delta_i > 0} \Delta_i \cdot \frac{\alpha T}{N} + o(1)$$

where  $\frac{\alpha T}{N}$  is the number of times each arm is pulled during exploration.

### 16.1.4 UCB1 Algorithm

The UCB1 algorithm is an alternative to the exploration-separate algorithm, which balances exploration and exploitation at each round. At time  $t$ , for each arm  $i$ , define

$$\text{UCB}_i(t) = \hat{\mu}_{i,t} + \sqrt{\frac{2 \log t}{N_{i,t}}}$$

where  $\hat{\mu}_{i,t}$  is the empirical mean and  $N_{i,t}$  is the number of times arm  $i$  has been pulled up to time  $t$ .

At each round, UCB1 selects the arm with the highest UCB value.

The instance dependent regret of UCB1 is bounded by

$$R_T^{(I.D.)}(\text{UCB1}) \leq \sum_{i: \Delta_i > 0} \left( \frac{8 \log T}{\Delta_i} + \Delta_i \right)$$

which is logarithmic in  $T$  for each suboptimal arm.

**17 October 13, 2025 (Midterm 2)**

**18 October 16, 2025 (Class 17)**

**19 October 20, 2025 (Class 18)**

**20 October 23, 2025 (Class 19)**

We have the instance independent regret of UCB1 algorithm as

$$R_T^{(I.I.)}(\text{UCB1}) = \mathcal{O} \left( \sqrt{NT \log T} \right)$$

**Proof of UCB1 instance independent regret bound:**

$$\begin{aligned} \text{event } E &:= \left\{ \left| \hat{\mu}_{i,t} - \mu_{i,t} \right| \leq \sqrt{\frac{2 \log T}{N_{i,t-1}}} = \varepsilon_{i,t} \quad \forall i, \forall t \leq T \right\} \\ \bar{E} &:= \exists t \leq T, \exists i, \text{ such that } E \text{ above is violated} \end{aligned}$$

We have

$$\begin{aligned} \Pr(\bar{E}) &\leq \frac{2}{T^4} TN, \quad \text{using union bound} \\ &= \frac{2N}{T^3} \leq \frac{2}{T^2} \quad (N < T) \end{aligned}$$

$$R_T^{(I.I.)}(\text{UCB1}) = \mathbb{E}[R_T | E] \Pr(E) + \mathbb{E}[R_T | \bar{E}] \Pr(\bar{E})$$

Let, without loss of generality, 1 be the best arm.

$$\Delta_{i_t} := \mu_1 - \mu_{i_t} \leq \mu_1 - \mu_{i_t} + \underbrace{\bar{\mu}_{i_t,t} - \bar{\mu}_{1,t}}_{\text{UCB}}$$

$$\mu_1 - \bar{\mu}_{1,t} = \mu_1 - \hat{\mu}_{1,N_{i_t,t-1}} - \sqrt{\frac{2 \log T}{N_{i_t,t-1}}} \leq 0$$

$$\bar{\mu}_{i_t,t} - \mu_{i_t} = \hat{\mu}_{i_t,N_{i_t,t-1}} + \sqrt{\frac{2 \log T}{N_{i_t,t-1}}} - \mu_{i_t} \leq 2\sqrt{\frac{2 \log T}{N_{i_t,t-1}}}$$

From the above 3 inequalities,

$$\Delta_{i_t} := \mu_1 - \mu_{i_t} \leq 2\sqrt{\frac{2 \log T}{N_{i_t,t-1}}}$$

Now,

$$\begin{aligned} \sum_{i:\Delta_i>0} \Delta_i \mathbb{E}[N_{i,T}] &= \mathbb{E} \left[ \sum_{i:\Delta_i>0} \Delta_i N_{i,T} \right] \\ &\leq \mathbb{E} \left[ \sum_{i:\Delta_i>0} 2\sqrt{\frac{2 \log T}{N_{i,T}}} N_{i,T} \right] \\ &= \mathbb{E} \left[ \sum_{i:\Delta_i>0} \sqrt{8 N_{i,T} \log T} \right] \\ &= \sum_{i:\Delta_i>0} \mathbb{E} \left[ \sqrt{8 N_{i,T} \log T} \right] \\ &= \sqrt{8 \log T} N \sum_{i:\Delta_i>0} \frac{1}{N} \mathbb{E} \left[ \sqrt{N_{i,t}} \right] \\ &\leq N \sqrt{8 \log T} \sqrt{\frac{T}{N}} = \sqrt{8 N T \log T} \\ &= \mathcal{O} \left( \sqrt{N T \log T} \right) \end{aligned}$$

### 20.0.1 Lower bound on regret for MAB setting

1. Instance independent:

$$\Omega(c\sqrt{NT})$$

2. Instance dependent:

$$\sum_{i:\Delta_i>0} \frac{\Delta_i \log T}{D_{KL}(\mu_i \parallel \mu^*)}$$

Some discussion on [Sub Gaussian Random Variables].

## 20.1 Chernoff bounding technique

Let  $\{X_i\}_{i=1}^n \approx \text{i.i.d.}$  be mean 0 random variables. Let  $\varepsilon, \delta > 0$  and

$$\Psi_X(\lambda) := \log(\mathbb{E}[\exp(\lambda X)]) .$$

Then,  $\exists T_0 > 0$  such that

$$Pr\left(\frac{1}{T} \sum_{t=1}^T X_t \geq \varepsilon\right) \leq \delta \quad \forall T \geq T_0.$$

**Proof of Chernoff bounding technique:**

$$\begin{aligned} Pr\left(\frac{1}{T} \sum_{t=1}^T X_t \geq \varepsilon\right) &= Pr\left(\sum_{t=1}^T X_t \geq T\varepsilon\right) \\ &= Pr\left(\exp\left(\sum_{t=1}^T \lambda X_t\right) \geq \exp(\lambda T\varepsilon)\right) \\ &\leq \mathbb{E}\left[\exp\left(\sum_{t=1}^T \lambda X_t\right)\right] \exp(-\lambda T\varepsilon) \\ &= \exp(-\lambda T\varepsilon) \prod_{t=1}^T \mathbb{E}[\exp(\lambda X_t)] \\ &= \exp(-\lambda T\varepsilon) \prod_{t=1}^T \exp(\Psi_X(\lambda)) \quad (i.i.d.) \\ &= \exp(-\lambda T\varepsilon) \exp(T\Psi_X(\lambda)) \\ &= \exp(-T(-\Psi_X(\lambda) + \lambda\varepsilon)) \end{aligned}$$

Define  $\Psi_\lambda^*(\varepsilon)$  as,

$$\Psi_\lambda^*(\varepsilon) := \sup_{\lambda > 0} [\lambda\varepsilon - \Psi_X(\lambda)]$$

and so

$$Pr\left(\frac{1}{T} \sum_{t=1}^T X_t \geq \varepsilon\right) \leq \exp(-T\Psi_\lambda^*(\varepsilon)) = \delta$$

## 21 October 27, 2025 (Class 20)

### 21.1 Thompson Sampling

$\alpha_{i_t}$  is equal  $S_{i_t} + 1$ ,  $\beta_{i_t}$  is equal  $N_{i_t} + 1$ , where S and N are success and instances when arm i is pulled.

**Algorithm : Thompson Sampling****Input:** Number of arms  $n$ **Initialize:**  $(Prior)Beta_i(1, 1)$  for all  $i \in [n]$ **for**  $t = 1, 2, \dots$  **do**- Draw  $\theta_i \sim Beta_i(\alpha_{i_t}, \beta_{i_t}) \quad \forall i \in [n]$ - Choose arm  $i_t = \arg \max_{i \in [n]}(\theta_i)$ - Observe  $r_{i_t, t} \sim Bernoulli(\mu_{i_t})$ 

- Update

$$\alpha_{i_t} \leftarrow \alpha_{i_t} + r_{i_t, t}$$

$$\beta_{i_t} \leftarrow \beta_{i_t} + 1$$

**end**

TODO: correct the above algorithm.

**21.1.1 Regret order of Thompson Sampling**

1. Instance dependent regret

$$\mathcal{O} \left( \sum_{i=1}^n \frac{\Delta_i (\log T + \log(\log T))}{D_{KL}(\mu_i \| \mu_{i^*})} \right)$$

2. Instance independent regret

$$\Omega \left( c\sqrt{NT} \right)$$

**21.2 Pure exploration setting**

Find the arm which gives the best expected reward.

1. Fixed confidence problem: Given  $\varepsilon, \delta > 0$ , goal is to find  $j$  such that the below probability is satisfied in least number of pulls.

$$Pr(\mu_j \geq \mu_{i^*} - \varepsilon) \geq 1 - \delta$$

2. Fixed budget problem: complementary to the above problem.

**21.3 Anytime confidence interval**

We have,

$$Pr \left( \bigcup_{t=1}^{\infty} |\hat{\mu}_{i_t, t} - \mu_{i_t}| \geq \sqrt{\frac{\log(4t^2/\delta)}{2t}} \right) \leq \delta$$

**Proof:** TODO: write proof

We also define

$$U(t, \delta) := \sqrt{\frac{\log(4t^2/\delta)}{2t}}$$



21.4 Successive elimination

21.5 See also

21.6 References